

IsoDecipher — Attractor Theory & Bifurcation Framework

Core Question

“Is APA switching a bifurcation event?”

When cells transition from naive → activated → terminal differentiated, does PUI (Proximal Usage Index) change gradually, or does it undergo a sudden, discontinuous jump — a bifurcation?

This reframes APA dynamics as a question in **dynamical systems theory**, connecting Waddington’s epigenetic landscape to single-cell multiomics.

Theoretical Foundation

Waddington Landscape as Potential Function

Cell state x evolves according to:

$$dx/dt = -\nabla U(x) + \eta(t)$$

Where:

- $U(x)$ = potential landscape (Waddington's "hills and valleys")
- $-\nabla U(x)$ = deterministic drift toward attractors
- $\eta(t)$ = stochastic noise (gene expression noise)

Key insight: RNA velocity estimates dx/dt directly. Therefore, we can reconstruct $\nabla U(x)$ from scVelo output.

Types of Cell States

State	Landscape Position	PUI Prediction	Entropy
Naive/stem	Shallow valley (metastable)	Intermediate, variable	High

Transitioning	Ridge/saddle point	Unstable, bimodal	Highest
Terminal differentiated	Deep valley (stable attractor)	Low variance, committed	Low
Cancer migrating (EMT)	Between attractors	Fluctuating	High
Proliferating	Outside attractor	High noise	High

Is APA Switching a Bifurcation?

What is a bifurcation?

A bifurcation occurs when a small change in a control parameter causes a qualitative change in system behavior.

In biology: a cell commits to one fate irreversibly. In PUI: a sudden shift from bimodal → unimodal distribution.

Types of bifurcation relevant to APA

Pitchfork bifurcation (cell fate decision):

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Before bifurcation: one stable PUI state (unimodal)
At bifurcation point: two equally stable PUI states (bimodal)
After bifurcation: one new stable PUI state (unimodal, committed)
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Saddle-node bifurcation (irreversible commitment):

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One attractor disappears → cell forced into new state
PUI jumps discontinuously
```

Algorithm: Detecting APA Bifurcation Along Pseudotime

Step 1: Compute PUI distribution shape along pseudotime

```
import numpy as np
import pandas as pd
from scipy.stats import kurtosis, skew
```

```

from scipy.signal import find_peaks

def pui_distribution_dynamics(adata, gene, n_bins=30, cells_per_bin=50):
    """
    Track the SHAPE of PUI distribution along pseudotime.
    Bifurcation signature: unimodal → bimodal → unimodal
    """
    pui_col = f'{gene}_PUI'
    pt = adata.obs['pseudotime'].values
    pui = adata.obs[pui_col].values

    # Sort by pseudotime
    order = np.argsort(pt)
    pt_sorted = pt[order]
    pui_sorted = pui[order]

    # Rolling window statistics
    results = []
    step = max(1, len(pt_sorted) // n_bins)

    for i in range(0, len(pt_sorted) - cells_per_bin, step):
        window_pui = pui_sorted[i:i+cells_per_bin]
        window_pt = pt_sorted[i:i+cells_per_bin]

        # Distribution shape metrics
        hist, edges = np.histogram(window_pui, bins=15, range=(0,1), density=True)
        peaks, _ = find_peaks(hist, height=0.1)

        results.append({
            'pseudotime': np.mean(window_pt),
            'mean_pui': np.mean(window_pui),
            'variance': np.var(window_pui),
            'bimodality': len(peaks), # 1=unimodal, 2=bimodal = bifurcation!
            'kurtosis': kurtosis(window_pui),
            'skewness': skew(window_pui),
            'entropy': -np.sum((hist/hist.sum()+1e-10) * np.log(hist/hist.sum()+1
        ))

    return pd.DataFrame(results)

# Bifurcation signature: bimodality peak in the middle of pseudotime
dyn = pui_distribution_dynamics(adata, 'IGHM')
print(dyn[['pseudotime', 'bimodality', 'variance', 'entropy']])

```

Step 2: Bimodality Coefficient — formal bifurcation test

```

def bimodality_coefficient(x):
    """
    BC > 0.555 suggests bimodal distribution (bifurcation point).
    BC = (skewness^2 + 1) / (kurtosis + 3*(n-1)^2/((n-2)*(n-3)))
    """
    n = len(x)
    s = skew(x)
    k = kurtosis(x)
    bc = (s**2 + 1) / (k + 3*(n-1)**2 / ((n-2)*(n-3)))
    return bc

def scan_bifurcation_pseudotime(adata, gene, n_bins=25, cells_per_bin=50):
    """
    Scan along pseudotime for bifurcation signature.
    Returns pseudotime point where bimodality is maximum.
    """
    pui_col = f'{gene}_PUI'
    pt = adata.obs['pseudotime'].values
    pui = adata.obs[pui_col].values

    order = np.argsort(pt)
    pt_s = pt[order]
    pui_s = pui[order]

    step = max(1, len(pt_s) // n_bins)
    bcs = []
    pts = []

    for i in range(0, len(pt_s) - cells_per_bin, step):
        window = pui_s[i:i+cells_per_bin]
        bc = bimodality_coefficient(window)
        bcs.append(bc)
        pts.append(np.mean(pt_s[i:i+cells_per_bin]))

    result = pd.DataFrame({'pseudotime': pts, 'bimodality_coeff': bcs})
    bifurcation_pt = result.loc[result['bimodality_coeff'].idxmax(), 'pseudotime']

    return result, bifurcation_pt

bc_traj, bifurcation_time = scan_bifurcation_pseudotime(adata, 'IGHM')
print(f"Bifurcation detected at pseudotime: {bifurcation_time:.3f}")

```

Step 3: Potential Landscape Reconstruction

```

def reconstruct_potential(adata, gene, n_bins=50):

```

```

"""
Reconstruct 1D potential landscape U(PUI) per pseudotime window.
 $U(x) = -\log(P(x))$  – deeper valleys = more stable states
"""

pui_col = f'{gene}_PUI'
pt = adata.obs['pseudotime'].values
pui = adata.obs[pui_col].values

# Split into early, mid, late pseudotime
tertiles = np.percentile(pt, [33, 67])

landscapes = {}
for label, mask in [
    ('early', pt < tertiles[0]),
    ('mid', (pt >= tertiles[0]) & (pt < tertiles[1])),
    ('late', pt >= tertiles[1])
]:
    pui_subset = pui[mask]
    hist, edges = np.histogram(pui_subset, bins=n_bins,
                               range=(0,1), density=True)

    hist = hist + 1e-10
    # Potential = -log(probability density)
    U = -np.log(hist)
    U = U - U.min() # normalize to 0
    bin_centers = (edges[:-1] + edges[1:]) / 2
    landscapes[label] = pd.DataFrame({'pui': bin_centers, 'potential': U})

return landscapes

landscapes = reconstruct_potential(adata, 'IGHM')

# Plot Waddington landscape evolution
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 3, figsize=(15, 4))
colors = {'early': 'steelblue', 'mid': 'darkorange', 'late': 'firebrick'}
titles = {'early': 'Early (Naive)', 'mid': 'Mid (Transitioning)', 'late': 'Late (

for ax, (stage, df) in zip(axes, landscapes.items()):
    ax.plot(df['pui'], df['potential'], color=colors[stage], linewidth=2.5)
    ax.fill_between(df['pui'], df['potential'], df['potential'].max(),
                    alpha=0.15, color=colors[stage])
    ax.set_xlabel('PUI (Proximal Usage Index)')
    ax.set_ylabel('Potential U(PUI)')
    ax.set_title(f'{titles[stage]}')
    ax.invert_yaxis() # Valleys at bottom, like Waddington

```

```
plt.suptitle('IGHM APA Landscape Evolution Along Differentiation',
             fontsize=13, fontweight='bold')
plt.tight_layout()
```

Step 4: Cell Stability Score

```
def cell_stability_score(adata, method='combined'):
    """
    Compute per-cell stability score.

    Low stability = proliferating, transitioning, or EMT cells
    High stability = terminal differentiated cells

    Methods:
    - 'velocity': use scVelo velocity magnitude (low = stable)
    - 'entropy': use transcriptional entropy (low = stable)
    - 'apa': use APA variance among neighbors (low = stable)
    - 'combined': weighted combination
    """
    scores = {}

    # 1. Velocity-based instability
    if 'velocity_length' in adata.obs.columns:
        vel_mag = adata.obs['velocity_length'].values
        scores['velocity_instability'] = vel_mag / vel_mag.max()

    # 2. Transcriptional entropy
    X = adata.X.toarray()
    X_norm = X / (X.sum(axis=1, keepdims=True) + 1e-10)
    from scipy.stats import entropy
    tx_entropy = np.apply_along_axis(entropy, 1, X_norm)
    scores['tx_entropy'] = tx_entropy / tx_entropy.max()

    # 3. APA-based stability (from IsoDecipher)
    pui_cols = [c for c in adata.obs.columns if c.endswith('_PUI')]
    if pui_cols:
        from sklearn.neighbors import NearestNeighbors
        pui_matrix = adata.obs[pui_cols].values

        nn = NearestNeighbors(n_neighbors=15)
        nn.fit(adata.obsm['X_umap'])
        distances, indices = nn.kneighbors(adata.obsm['X_umap'])

        apa_instability = []
```

```

for i in range(adata.n_obs):
    neighbor_pui = pui_matrix[indices[i]]
    local_var = np.nanmean(np.nanvar(neighbor_pui, axis=0))
    apa_instability.append(local_var)

apa_instability = np.array(apa_instability)
scores['apa_instability'] = apa_instability / (apa_instability.max() + 1e

# Combined score
score_matrix = np.column_stack(list(scores.values()))
combined = np.nanmean(score_matrix, axis=1)

adata.obs['stability_score'] = 1 - combined # flip: high = stable
adata.obs['instability_score'] = combined

for key, val in scores.items():
    adata.obs[key] = val

return adata

```

Biological Predictions

For B cell → Plasma cell differentiation:

Cell State	Expected PUI (IGHM)	Stability	Bifurcation?
Naïve B cell	~0.5 (mixed)	Medium	No
GC B cell	Variable, bimodal	Low (transitioning)	YES
Plasmablast	Shifting toward G0	Medium	No
Long-lived plasma cell	~0.9 (secreted, G0)	High	No

For Cancer (EMT):

Cell State	Expected PUI (CD44)	Stability
Epithelial	Low variance	High
Hybrid EMT	Bimodal, high variance	Lowest
Mesenchymal/migrating	Shifted, low variance	Medium

Publication Roadmap

Paper title options:

- 1. *"APA switching is a bifurcation event in immune cell differentiation: single-cell evidence from IsoDecipher"*
- 2. *"Reconstructing the Waddington landscape from APA dynamics in single-cell RNA-seq data"*
- 3. *"Information-theoretic signatures of cell state transitions revealed by targeted isoform quantification"*

Key figures:

- 1. Waddington landscape evolution (early/mid/late)
 - 2. Bimodality coefficient along pseudotime (bifurcation point)
 - 3. Cell stability score UMAP (proliferating vs terminal)
 - 4. APA trajectory with bifurcation annotation
 - 5. Validation: EMT genes in cancer cell line data
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Connection to Your Existing Work

Your background	Connection to this framework
ODE modeling (Nat Commun 2022)	Attractor theory IS ODE theory — same math
SEC-seq secretome	Secretion switching = bifurcation in output modality
IsoDecipher APA	PUI trajectory = path through landscape
RNA velocity	dx/dt = gradient of potential landscape
Math modeling coursework	Attractor, bifurcation = core concepts you already know
Flow cytometry	Validate predicted cell states experimentally

Next Steps (週一開始)

1. **PUI notebook** — basic downstream analysis (1 day)
2. **APA trajectory** — pseudotime + smoothing (1 day)
3. **Bifurcation detection** — bimodality coefficient (1 day)
4. **Waddington landscape plot** — most visual, most impressive (1 day)
5. **Cell stability score** — combine velocity + entropy + APA (2 days)
6. **Preprint draft** — bioRxiv submission (2 weeks)